Generalization theorem for the axiomatic calculus

The Open Logic Project contributors

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-first-order-logic-axiomatic-deduction/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: in-source axiom schemes Ax2/Ax3 from the quantifier axiom group.

Editorial note. The label thm:Generalization names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own thm environment in the unit source.

Allowed standard prerequisites (editorial).

- Axiom schemes Ax2 and Ax3. Same unit, quantifier axiom group (Ax2 distribution, Ax3 universal instantiation). Basis case Ax3 for members of Gamma; inductive step Ax2 for the MP case.

Theorem 0.1 (Generalization)**.** *If $\Gamma \vdash A$ and x is not free in any formula in Γ, then $\Gamma \vdash \forall x\, A$.*

Proof. By induction on Thm_Γ: if A is an axiom, so is $\forall x\, A$. If $A \in \Gamma$ then x is not free in A, so that $A \to \forall x\, A$ is an axiom (**Ax3**), and $\Gamma \vdash \forall x\, A$ by MP. Now suppose A follows by MP because $\Gamma \vdash B$ and $\Gamma \vdash B \to A$. By induction hypothesis, $\Gamma \vdash \forall x\, B$ and $\Gamma \vdash \forall x\, (B \to A)$. But by **Ax2** also

$$\Gamma \vdash \forall x\, (B \to A) \to (\forall x\, B \to \forall x\, A),$$

and two applications of MP give $\Gamma \vdash \forall x\, A$ as desired. □